

Week-8

1) For the following sequence of pages using FIFO, LRU, MRU & optimal page replacement Algorithms.

Sequence : 7 0 1 2 0 3 0 4 2 3 0 3 0 3 2 1

No. of frames in physical memory are : 4

Given,

Reference string: 7 0 1 2 0 3 0 4 2 3 0 3 0 3 2 1

No. of frames : 4

(i) FIFO - first in first out

In FIFO, we replace the page that entered the memory first.

step	page	f ₁	f ₂	f ₃	f ₄	Result
1	7	7	-	-	-	F
2	0	7	0	-	-	F
3	2	7	0	2	-	F
4	0	7	0	2	0	H
5	3	7	0	2	3	F
6	0	7	0	2	3	H
7	4	4	0	2	3	F
8	2	4	0	2	3	H
9	3	4	0	2	3	H
10	0	4	0	2	3	H

11	3	4	0	2	3	H
12	2	4	0	2	3	H
13	1	1	0	2	3	F

Page faults = 6

Page hits = 7

(ii) LRU - Least Recently used

In LRU, we replace the page that has not been used for the longest time.

Step	page	f ₁	f ₂	f ₃	f ₄	Result
1	7	7	0	0	0	F
2	0	7	0	0	0	F
3	2	7	0	0	2	F
4	0	7	0	2	0	H
5	3	7	0	2	3	F
6	0	7	0	2	3	H
7	4	4	0	2	3	F
8	2	4	0	2	3	H
9	3	4	0	2	3	H
10	0	4	0	2	3	H
11	3	4	0	2	3	H
12	2	4	0	2	3	H
13	1	4	0	2	1	F

Page-faults = 6

Page hits = 7

(iii) MRU - Most Recently used

In MRU, we Replace the page that was used most recently.

Step	page	f ₁	f ₂	f ₃	f ₄	Result
1	7	1	0	0	0	F
2	0	0	1	0	0	F
3	3	0	0	1	0	F
4	0	0	1	0	0	H
5	3	0	0	1	0	F
6	0	0	1	0	0	H
7	4	0	0	0	1	F
8	2	0	0	0	0	H
9	3	0	0	0	1	H
10	0	0	1	0	0	F
11	3	0	0	1	0	F
12	2	0	1	0	0	H
13	1	1	0	0	0	H

Page-faults = 8

Page hits = 5

(iv) optimal page replacement

In optimal page, replace the page whose next use is farthest in the future.

Step	page	f_1	f_2	f_3	f_4	Result
1	7	7	-	-	-	F
2	0	7	0	-	-	F
3	2	7	0	2	-	F
4	0	7	0	2	-	H
5	3	7	0	2	3	F
6	0	7	0	2	3	H
7	4	7	0	2	4	F
8	2	7	0	2	4	H
9	13	7	0	2	4	H
10	0	7	0	2	4	H
11	3	7	0	2	4	H
12	2	7	0	2	4	H
13	1	7	0	2	4	F

Page faults = 6

Page hits = 7

Therefore, total reference string = 13 page

So,

FIFO = 6 faults
LRU = 6 faults
MRU = 8 faults
optimal = 6 faults

2) Apply -FIFO page replacement algorithm for the following sequence of pages and check whether the belady's / Anamoly exists or not for checking up belady Anamoly take the physical memory size as 3 frames and 4 frames

Page Sequence : 1 2 3 4 1 2 5 1 2 3 4 5

we need to check with 3 frames and 4 frames:

* with 3 frames

1 2 3 → fault

4 replaces 1 → fault

1 Replaces 2 → fault

2 replaces 3 → fault

5 replaces 4 → fault

1 → hit

2 → hit

3 replaces 1 $\rightarrow$ fault

4 replaces 2 $\rightarrow$ fault

5 $\rightarrow$ hit

Total page faults = 9

* with 4 frames

1 2 3 4 $\rightarrow$ faults

1 $\rightarrow$ hit

2 $\rightarrow$ hit

5 replaces 1 $\rightarrow$ fault

1 replaces 2 $\rightarrow$ fault

2 replaces 3 $\rightarrow$ fault

3 replaces 4 $\rightarrow$ fault

4 replaces 5 $\rightarrow$ fault

5 replaces 1 $\rightarrow$ fault

Total page faults = 10

(frames page #)

frames fifo page faults.

3 frames 9

4 frames 10

Since page faults increase from 9 $\rightarrow$ 10 when frames increase from 3 $\rightarrow$ 4, Belady's Anomaly also exists.